Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Let $\mathrm{A}=\{\mathrm{p}, \mathrm{q}, \mathrm{r}\}$. Which of the following is an equivalence relation in A ?

(A) $\mathrm{R}_1=\{(\mathrm{p}, \mathrm{q}),(\mathrm{q}, \mathrm{r}),(\mathrm{p}, \mathrm{r}),(\mathrm{p}, \mathrm{q})\}$

(B) $R_2=\{(r, q),(r, p),(r, r),(q, q)\}$

(C) $\mathrm{R}_3^2=\{(\mathrm{p}, \mathrm{p}),(\mathrm{q}, \mathrm{q}),(\mathrm{r}, \mathrm{r}),(\mathrm{p}, \mathrm{q})\}$

(D) None of these

Q.2

Let $\mathrm{R}=\{(1,3),(4,2),(2,4),(2,3),(3,1)\}$ be relation on the set $\mathrm{A}=\{1,2,3,4\} \ldots$ The relation R is

(A) reflexive

(B) transitive

(C) not symmetric

(D) a function

Q.3

Let $P=\left\{(x, y):\left|x^2+y^2\right|=1, x, y \in R\right\}$. Then Pis

(A) Reflexive

(B) Symmetric

(C) Transitive

(D) Anti-symmetric

Q.4

Let ρ be the relation on the set R of all real numbers defined by setting aρb iff $|\mathrm{a}-\mathrm{b}| \leq \frac{1}{2}$. Then, ρ is

(A) reflexive and symmetric but not transitive

(B) symmetric and transitive but not reflexive

(C) transitive but neither reflexive nor symmetric

(D) None of these

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Q.5

Let R be a relation over the $\mathrm{N} \times \mathrm{N}$ and it is defined by $(a, b)\mathrm{R}(c, d) \Rightarrow a+d=b+c$. Then, R is

(A) Reflexive only

(B) Symmetric only

(C) Transitive only

(D) An equivalence relation

Q.6

R is a relation from $\{11, 12, 13\}$ to $\{8, 10, 12\}$ defined by $y=x-3$. The relation R^{-1} is

(A) $\{(11, 8), (13, 10)\}$

(B) $\{(8, 11), (10, 13)\}$

(C) $\{(8, 11), (9, 12), (10, 13)\}$

(D) None of these

Q.7

Let $\mathrm{R}=\{(x, y) : x, y \in N$ and $x^2-4xy+3y^2=0\}$
where N is the set of all natural numbers. Then the relation R is :

(A) reflexive but neither symmetric nor transitive.

(B) symmetric and transitive.

(C) reflexive and symmetric.

(D) reflexive and transitive.

Q.8

Let n be a fixed positive integer. Define a relation R in the set Z of integers by aRb if and only if $\frac{\mathrm{n}}{\mathrm{a}-\mathrm{b}}$. The relation R is

(A) reflexive

(B) symmetric

(C) transitive

(D) an equivalence relation

Q.9

Let $n(A)=4$ and $n(B)=6$. The number of one to one functions from A to B is

(A) 24

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(B) 60

(C) 120

(D) 360

Q.10

The relation R defined in $\mathrm{A}=\{1,2,3\}$ by aRb, if $\left|\ a^2-b^2\ \right| \leq 5$. Which of the following is false?

(A) $\mathrm{R}=\{(1,1),(2,2),(3,3),(2,1),(1,2),(2,3)$ $(3,2)\}$

(B) $\mathrm{R}^{-1}=\mathrm{R}$

(C) Domain of $\mathrm{R}=\{1,2,3\}$

(D) Range of $\mathrm{R}=\{5\}$

Q.11

Total number of equivalence relations defined in the set $\mathrm{S}=\{\mathrm{a}, \mathrm{b}, \mathrm{c}\}$ is :

(A) 5

(B) $3!$

(C) 2^3

(D) 3^3

Q.12

Let a relation R in the set R of real numbers be defined as $(a, b) \in R$ if and only if $1+ab > o$ for all $a, b \in R$

The relation R is

(A) reflexive and symmetric

(B) symmetric and transitive

(C) only transitive

(D) an equivalence relation

Q.13

Let R be a relation on $\mathbf{N} \times \mathbf{N}$ defined by $(a, b) R(c, d) \Rightarrow ad(b+c)=bc(a+d). R$ is

(A) a partial order relation

(B) an equivalence relation

(C) an identity relation

(D) None of these

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Q.14

Let R and S be two non-void relations in a set A. Which of the following statements is not true?

(A) R and S transitive $\Rightarrow \mathrm{R} \cup \mathrm{S}$ is transitive

(B) R and S transitive $\Rightarrow \mathrm{R} \cap \mathrm{S}$ is transitive

(C) R and S symmetric $\Rightarrow \mathrm{R} \cup \mathrm{S}$ is symmetric

(D) R and S reflexive $\Rightarrow \mathrm{R} \cap \mathrm{S}$ is reflexive

Q.15

Let S be any set and $P(S)$ be its power set. We define a relation R on $\mathrm{P}(\mathrm{S})$ by ARB to mean $\mathrm{A} \subseteq \mathrm{B}$; $\forall \mathrm{A}, \mathrm{B} \in \mathrm{P}(\mathrm{S})$. Then R is

(A) equivalence relation

(B) not an equivalence but partial order relation

(C) both equivalence and partial order relation

(D) None of these

Answers & Solutions

Q.1 Answer:

None of these

Solution:

R_1 **is not reflexive, because** $(\mathrm{q}, \mathrm{q})(\mathrm{r}, \mathrm{r}) \notin \mathrm{R}_1$
$\therefore \mathrm{R}_1$ **is not an equivalence relation** R_2 **is not reflexive, because** $(p, p) \notin R_2 \therefore \mathrm{R}_2$ **is not an equivalence relation.** R_3 **is reflexive, because** $(\mathrm{p}, \mathrm{p}), (\mathrm{q}, \mathrm{q}), (\mathrm{r}, \mathrm{r}) \in \mathrm{R}_3$.
R_3 **is not symmetric, because** $(\mathrm{p}, \mathrm{q}) \in \mathrm{R}_3$
but $(\mathrm{q}, \mathrm{p}) \notin \mathrm{R}_3$.

Q.2 Answer:

not symmetric

Solution:

$(2, 3) \in \mathrm{R}$ **but** $(3, 2) \notin \mathrm{R}$
$\therefore \quad \mathrm{R}$ **is not symmetric**

Q.3 Answer:

Symmetric

Solution:

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Obviously, the relation is not reflexive and transitive but it is symmetric, because $x^2 + y^2 = 1 \Rightarrow y^2 + x^2 = 1$

Q.4 Answer:

reflexive and symmetric but not transitive

Solution:

ρ is reflexive, since $|\mathrm{a} - \mathrm{a}| = 0 < \frac{1}{2}$ for all $\mathrm{a} \in \mathrm{R}$.
ρ is symmetric, since $\Rightarrow |\mathrm{b} - \mathrm{a}| < \frac{1}{2}$
ρ is not transitive. For, if we take three numbers $\frac{3}{4}, \frac{1}{3}, \frac{1}{8}$.
Then, $\left|\frac{3}{4} - \frac{1}{3}\right| = \frac{5}{12} < \frac{1}{2}$ and $\left|\frac{1}{3} - \frac{1}{8}\right| = \frac{5}{24} < \frac{1}{2}$
But, $\left|\frac{3}{4} - \frac{1}{8}\right| = \frac{5}{8} > \frac{1}{2}$
Thus, $\frac{3}{4}\rho\frac{1}{3}$ and $\frac{1}{3}\rho\frac{1}{8}$ but $\frac{3}{4}(\sim \rho)\frac{1}{8}$

Q.5 Answer:

An equivalence relation

Solution:

We have $(a, b)\mathrm{R}(a, b)$ for all $(a, b) \in \mathrm{N} \times \mathrm{N}$
As $a + b = b + a$. Hence, R is reflexive R is symmetric for we have $(a, b)\mathrm{R}(c, d) \Rightarrow a + d = b + c \quad \Rightarrow d + a = c + b$
$\Rightarrow c + b = d + a \quad \Rightarrow (c, d)\mathrm{R}(e, f)$
Then, by definition of R, we have $a + d = b + c$ and $c + f = d + e$
So, by addition, we get $a + d + c + f = b + c + d + c$ or $a + f = b + e$
Hence, $(\mathrm{a}, \mathrm{b})\mathrm{R}(\mathrm{e}, \mathrm{f})$
Thus, $(a, b)R(c, d)$ and $(c, d)R(e, f)$
$\Rightarrow (\mathrm{a}, \mathrm{b})\mathrm{R}(\mathrm{e}, \mathrm{f})$

Q.6 Answer:

$\{(8, 11), (10, 13)\}$

Solution:

We have, $\mathrm{R} = \{(\mathrm{x}, \mathrm{y}) : \mathrm{y} = \mathrm{x} - 3, \mathrm{x} = 11$ or 12 or 13, $\mathrm{y} = 8$ or 10 or 12$\} = \{(11, 8), (13, 10)\} \therefore \mathrm{R}^{-1}\{(\mathrm{y}, \mathrm{x}) : (\mathrm{x}, \mathrm{y}) \in \mathrm{R}\} = \{(8, 11), (10, 13)\}$

Q.7 Answer:

reflexive but neither symmetric nor transitive.

Solution:

$\mathrm{R} = \{(x, y) : x, y \in \mathrm{N}$ **and** $x^2 - 4xy + 3y^2 = 0\}$
Now,

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$x^2 - 4xy + 3y^2 = 0$
$\Rightarrow (x - y)(x - 3y) = 0$
$\therefore x = y$ or $x = 3y$
$\therefore \mathrm{R} = \{(1, 1), (3, 1), (2, 2), (6, 2), (3, 3), (9, 3), \ldots\ldots\}$

Since $(1, 1), (2, 2), (3, 3), \ldots\ldots$ are present in the relation, therefore R is reflexive.

Since $(3, 1)$ is an element of R but $(1, 3)$ is not the element of R, therefore R is not symmetric.

As $x = 3y$, we consider two elements:
1. $x = 3, y = 1$ **and**
2. $x = 9, y = 3$

Now, the elements can be written as $(3, 1)$ and $(9, 3)$
Therefore, as $(9, 3)$ and $(3, 1)$ is present, there must also be $(9, 1)$. But it is not there. Therefore, the R is not transitive.

Q.8 Answer:

an equivalence relation

Solution:

We have, $R = \{(a, b) : n/(a - b) : a, b \in Z\}$
Let $z \in Z. \therefore a - a = 0 = n \times 0$ **ie,** $n/(a - a)$
$aRa \Rightarrow (a - a)$ **divisible by** n**, which is true for any integer 'a 'as 0 divisible by** n
$\therefore (a, a) \in R$ **ie,** R **is reflexive.**
Let $(a, b) \in R.$
$\therefore n/(a - b)$
$\Rightarrow \mathrm{n}/-(\mathrm{a} - \mathrm{b}) \Rightarrow \mathrm{n}/(\mathrm{b} - \mathrm{a}) \Rightarrow (\mathrm{b}, \mathrm{a}) \in \mathrm{R}$
$\therefore \mathrm{R}$ **is symmetric.**
Let $(a, b), (b, c) \in R.$
$\therefore n/(a - b), n/(b - c)$
$\Rightarrow \mathrm{n}/(\mathrm{a} - \mathrm{b}) + (\mathrm{b} - \mathrm{c}) \Rightarrow \mathrm{n}/\mathrm{a} - \mathrm{c} \Rightarrow (\mathrm{a}, \mathrm{c}) \in \mathrm{R}$
$\therefore \mathrm{R}$ **is transitive.**
$\therefore \mathrm{R}$ **is an equivalence relation.**

Q.9 Answer:

360

Solution:

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Required number of one to one functions

$$= {}^{6}\mathrm{P}_{4}$$
$$= 6 \times 5 \times 4 \times 3 = 360$$

Q.10 Answer:

Range of $\mathrm{R} = \{5\}$

Solution:

$\mathrm{R} = \{(1,1),(1,2),(2,1),(2,2),(2,3),(3,2),(3,3)\}$
$\mathrm{R}^{-1} = \{(\mathrm{y},\mathrm{x}) : (\mathrm{x},\mathrm{y}) \in \mathrm{R}\}$
$= \{(1,1),(2,1),(1,2),(2,2),(3,2),(2,3),(3,3)\} = R$
Domain of $R = \{x : (x,y) \in R\} = \{1,2,3\}$
Range of $\mathrm{R} = \{\mathrm{y} : (\mathrm{x},\mathrm{y}) \in \mathrm{R}\} = \{1,2,3\}$

Q.11 Answer:

5

Solution:

The smallest equivalence relation is the identity relation $\mathrm{R}_1 = \{(\mathrm{a},\mathrm{a}),(\mathrm{b},\mathrm{b}),(\mathrm{c},\mathrm{c})$
Then two ordered pairs of two distinct elements can be added to give three more equivalence relations $R_2 = \{(a,a),(b,b),(c,c),(a,b),(b,a)\}$
Similarly R_3 **and** R_4**. Finally the largest equivalence relation i.e, the universal relation**

Q.12 Answer:

reflexive and symmetric

Solution:

We have, $\mathrm{R} = \{(\mathrm{a},\mathrm{b}) : 1 + \mathrm{ab} > 0, \mathrm{ab} \in \mathrm{R}\}$
Let $a \in R, \therefore a^2 \geq 0$ **or** $1 + a^2 > 0$ **or** $(a,a) \in R$
$\therefore \mathrm{R}$ **is reflexive. Let** $(a,b) \in R. \Rightarrow 1 + ab > 0 \Rightarrow 1 + ba > 0$
$\Rightarrow (\mathrm{b},\mathrm{a}) \in \mathrm{R} \therefore \mathrm{R}$ **is symmetric.**
$\left(2, \frac{1}{3}\right) \in \mathrm{R}$ **because** $1 + 2\left(\frac{1}{3}\right) = \frac{5}{3} > 0$
$\left(\frac{1}{3}, -1\right) \in \mathrm{R}$ **because** $1 + \frac{1}{3}(-1) = \frac{2}{3} > 0$
Now, $(2,-1) \in \mathrm{R}$ **if** $1 + 2(-1) = -1 < 0$**, which is not true.** R **is not transitive**

Q.13 Answer:

an equivalence relation

Solution:

We observe the following properties:
Reflexivity : Let (a,b) **be an arbitrary element of** $\mathbf{N} \times \mathbf{N}$

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Then, $(a, b) \in \mathbf{N} \times \mathbf{N} \Rightarrow a, b \in \mathbf{N}$

$\Rightarrow ab(b+a) = ba(a+b) \Rightarrow (a,b)R(a,b)$

Thus, $(a,b)R(a,b)$ **for all** $(a,b) \in \mathbf{N} \times \mathbf{N}$

So, R **is reflexive on** $\mathbf{N} \times \mathbf{N}$ **Symmetry : Let** $(a, b), (c, d) \in \mathbf{N} \times \mathbf{N}$ **be such that** $(a,b)R(c,d)$**. Then,** $(a,b)R(c,d)$

$\Rightarrow ad(b+c) = bc(a+d)$

$\Rightarrow cb(d+a) = da(c+b) \Rightarrow (c,d)R(a,b)$

Thus, $(a,b)R(c,d) \Rightarrow (c,d)R(a,b)$ **for all** $(a,b), (c,d) \in \mathbf{N} \times \mathbf{N}$

So, R **is symmetric on** $\mathbf{N} \times \mathbf{N}$**. Transitivity : Let** $(a, b), (c, d), (e, f) \in \mathbf{N} \times \mathbf{N}$ **such that** $(a, b)R(c, d)$ **and** $(c, d)R(e, f)$

Then, $(a,b)R(c,d) \Rightarrow ad(b+c) = bc(a+d)$

$\Rightarrow \frac{b+c}{bc} = \frac{a+d}{ad} \Rightarrow \frac{1}{b} + \frac{1}{c} = \frac{1}{a} + \frac{1}{d}$ **...(i)**

and, $(c,d)R(e,f) \Rightarrow cf(d+e) = de(c+f)$

$\Rightarrow \frac{d+e}{de} = \frac{c+f}{cf} \Rightarrow \frac{1}{d} + \frac{1}{e} = \frac{1}{c} + \frac{1}{f}$ **...(ii)**

Adding (i) and (ii), we get $\left(\frac{1}{b} + \frac{1}{c}\right) + \left(\frac{1}{d} + \frac{1}{e}\right) = \left(\frac{1}{a} + \frac{1}{d}\right) + \left(\frac{1}{c} + \frac{1}{f}\right)$

$\Rightarrow \frac{1}{b} + \frac{1}{e} = \frac{1}{a} + \frac{1}{f} \Rightarrow \frac{b+e}{be} = \frac{a+f}{af}$

$\Rightarrow af(b+e) = be(a,f) \Rightarrow (a,b)R(e,f)$

Thus, $(a, b)R(c, d)$ **and** $(c, d)R(e, f)$

$\Rightarrow (a, b)R(e, f)$ **for all** $(a, b), (c, d), (e, f) \in N \times N$

So, R **is transitive on** $N \times N$ **Hence,** R **being reflexive, symmetric and transitive, is an equivalence relation on** $N \times N$

Q.14 Answer:

R and S transitive $\Rightarrow R \cup S$ is transitive

Solution:

Let $(a, b), (b, c) \in R \cup S$

It is possible that $(a,b) \in R - S$ **and** $(b,c) \in S - R$ **In such a case, we cannot say that** $(a,c) \in R$ **or** $(a,c) \in S$

$\therefore (a, c)$ **may not be in** $R \cup S$**.**

$\therefore R \cup S$ **is not transitive.**

(b) Let $(a,b), (b,c) \in R \cap S$

$\therefore (a,b), (b,c) \in Rand(a,b), (b,c) \in S$

$\therefore (a,c) \in R$ **and** $(a,c) \in S$

$\therefore (a,c) \in R \cap S$

$\therefore R \cap S$ **is transitive.**

(c) Let $(a,b) \in R \cup S \quad \therefore (a,b) \in R$ **or** $(a,b) \in S$

Now, $(a,b) \in R \Rightarrow (b,a) \in R (\because R$ **is symmetric)**

$(a,b) \in S \Rightarrow (b,a) \in S \quad (\because S$ **is symmetic** $)$

$\therefore (b, a) \in R \cup S \therefore R \cup S$ **is symmetric.**

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(d) Let $a \in A$. $\therefore \quad (a, a) \in R$ and $(a, a) \in S$.
$\therefore (a, a) \in R \cap S. \quad \therefore R \cap S$ is reflexive.

Q.15 Answer:

not an equivalence but partial order relation

Solution:

(i) $A \subseteq A$ie, $ARA, \forall A \in P(S)$
$\therefore R$ is reflexive.
(ii) $A \subseteq B \not B \subseteq A$
$\therefore$ ARB $\neq$ BRA. So, R is not symmetric.
(iii) ARB and $BRA \Rightarrow A \subseteq B$ and $B \subseteq A \Rightarrow A = B$.
Thus, R is anti-symmetric.
(iv) ARB and $BRC \quad \Rightarrow A \subseteq B$ and $B \subseteq C$
$\Rightarrow A \subseteq C \Rightarrow ARC$
$\therefore R$ is transitive relation. Thus, R is partially ordered relation but not an equivalence relation.

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